

CHYULBYU, Russian Federation

WMO: 311230

Lat: 57.816N

Lon: 130.900E

Elev: 255

StdP: 98.30

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.5	-43.9	-47.4	0.0	-45.2	-45.8	0.0	-43.8	4.3	-27.7	3.5	-27.6	0.2	170	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	30.1	19.2	27.8	18.1	25.5	17.5	20.8	26.7	19.6	25.4	18.5	24.0	1.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.0	23.3	17.7	13.1	22.3	16.5	12.1	20.9	61.5	26.8	57.6	25.7	53.8	24.0	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.3	2.7	2.3	DB	-46.9	33.5	1.8	2.0	-48.1	34.9	-49.2	36.1	-50.2	37.2	-51.5	38.6	(4)
(5)			WB	-46.8	23.5	1.8	1.1	-48.1	24.4	-49.2	25.0	-50.2	25.7	-51.5	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-6.7	-34.1	-28.8	-16.2	-3.0	7.7	15.0	18.0	15.0	6.8	-5.2	-22.9	-33.9	(6)	
	DBStd	19.63	5.51	5.53	6.57	5.70	3.85	3.51	3.14	3.67	4.14	6.04	7.70	5.69	(7)	
	HDD10.0	6688	1366	1087	813	389	91	4	0	6	114	471	988	1361	(8)	
	HDD18.3	9182	1624	1320	1072	639	328	107	45	113	347	729	1238	1619	(9)	
	CDD10.0	602	0	0	0	0	21	156	248	161	17	0	0	0	(10)	
	CDD18.3	54	0	0	0	0	0	8	35	11	0	0	0	0	(11)	
	CDH23.3	998	0	0	0	0	22	295	492	188	1	0	0	0	(12)	
	CDH26.7	310	0	0	0	0	4	86	173	48	0	0	0	0	(13)	
(14) Wind	WSAvg	0.8	0.7	0.6	0.7	1.0	0.9	0.8	0.7	0.7	0.8	0.9	0.8	0.7	(14)	
(15) Precipitation	PrecAvg	391	11	6	7	12	41	55	72	75	57	26	16	11	(15)	
	PrecMax	600	25	14	17	33	78	109	135	154	137	93	58	24	(16)	
	PrecMin	273	4	1	2	2	13	24	15	13	20	4	4	6	(17)	
	PrecStd	74	6	3	5	9	17	24	31	32	25	17	11	4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-18.2	-10.5	5.5	16.8	26.1	32.0	33.2	30.9	22.1	10.4	-1.0	-15.4	(19)	
		MCWB	-18.5	-11.7	0.9	7.8	13.9	18.8	20.1	20.5	14.9	5.6	-2.5	-15.8	(20)	
	2%	DB	-21.5	-14.4	1.5	12.3	22.7	29.2	30.8	27.7	19.0	6.6	-6.3	-20.3	(21)	
		MCWB	-21.8	-15.1	-1.8	5.1	12.1	17.2	20.0	19.2	13.3	3.5	-7.2	-20.6	(22)	
	5%	DB	-23.7	-17.7	-1.4	9.4	20.1	27.0	28.5	25.2	16.4	4.5	-9.1	-22.9	(23)	
		MCWB	-23.9	-18.3	-4.1	3.5	11.0	16.2	19.3	18.1	11.7	2.1	-9.8	-23.2	(24)	
	10%	DB	-25.7	-20.3	-4.2	6.8	17.3	24.6	25.9	22.8	13.9	2.3	-11.8	-25.6	(25)	
		MCWB	-26.0	-20.8	-6.4	1.8	9.5	15.6	18.6	17.1	10.2	0.4	-12.4	-25.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-18.5	-11.5	1.0	8.2	14.7	20.5	22.8	21.6	16.0	6.1	-2.7	-15.8	(27)	
		MCDB	-18.3	-10.5	5.0	16.8	23.5	27.0	29.3	27.5	20.7	9.1	-1.0	-15.5	(28)	
	2%	WB	-21.8	-15.1	-1.7	5.4	12.8	18.9	21.4	20.1	14.0	4.0	-7.0	-20.6	(29)	
		MCDB	-21.5	-14.4	1.4	11.6	20.6	26.2	27.7	25.6	17.8	6.1	-6.2	-20.3	(30)	
	5%	WB	-23.9	-18.3	-3.9	3.7	11.5	17.7	20.2	18.9	12.4	2.2	-9.7	-23.2	(31)	
		MCDB	-23.6	-17.7	-1.4	8.9	18.8	24.2	26.1	24.1	15.6	4.2	-9.2	-22.9	(32)	
	10%	WB	-25.9	-20.8	-6.4	2.2	10.1	16.6	19.3	17.7	10.9	0.5	-12.4	-25.8	(33)	
		MCDB	-25.7	-20.3	-4.2	6.4	16.4	22.8	24.9	22.0	13.4	2.1	-11.8	-25.6	(34)	
(35) Mean Daily Temperature Range	MDBR		7.9	12.8	17.6	15.8	14.6	15.8	14.0	13.5	11.4	9.4	8.5	6.3	(35)	
	5% DB	MCDBR	8.5	14.3	18.8	17.7	21.2	21.6	19.5	18.7	15.7	10.0	8.8	7.0	(36)	
		MCWBR	8.3	13.5	15.3	10.8	11.3	10.5	9.7	10.8	9.9	6.8	7.9	6.9	(37)	
	5% WB	MCDBR	8.4	14.1	18.4	16.9	18.9	17.4	16.0	16.5	13.6	9.5	8.8	7.1	(38)	
		MCWBR	8.2	13.3	15.0	10.6	10.3	9.5	8.8	10.0	9.0	6.7	7.9	6.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.241	0.278	0.319	0.402	0.462	0.444	0.484	0.417	0.347	0.310	0.254	0.226	(40)	
	taud		2.042	2.048	2.004	1.886	1.980	2.186	2.090	2.280	2.409	2.220	2.045	1.981	(41)	
	Ebn at Noon		599	743	809	790	770	796	749	781	794	701	564	485	(42)	
	Edh at Noon		55	88	123	166	162	134	145	112	84	74	54	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.71	1.78	3.44	5.15	5.00	5.55	5.14	4.08	2.61	1.50	0.91	0.47	(44)	
	RadStd		0.04	0.09	0.13	0.34	0.41	0.39	0.38	0.35	0.20	0.16	0.05	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page